



MERGE 4 DESIGN LLC

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

PROFESSIONAL FEE PROPOSAL

1212 Lobdel Avenue — Multi-Unit Affordable Infill Housing

Wilmington, DE · Predevelopment & Architecture Services · April 2026

Submitted To	Wilmington Housing Authority (WHA) · Wilmington, DE
Submitted By	Merge 4 Design LLC (M4D) · 500 Delaware Ave., Suite 1 #1551, Wilmington, DE 19899 · 302 367 5726 · merge4design.com
Date Submitted	April 2026
Project Reference	M2_2026_012 · 1212 Lobdel Avenue
Proposal Valid	90 days from submission date



MERGE 4 DESIGN LLC

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

01. Project Identification

Project Name	1212 Lobdel Avenue — Multi-Unit Affordable Infill Housing
Parcel ID	2605130241
Address	1212 Lobdel Avenue, Wilmington, DE (City of Wilmington limits)
Client / Owner	Wilmington Housing Authority (WHA)
Project Lead	Merge 4 Design LLC (M4D)
Zoning Jurisdiction	City of Wilmington Zoning Ordinance, Chapter 48 — R-3 District
M4D Project Number	M2_2026_005 · April 2026



MERGE 4 DESIGN LLC

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

02. Project Context & Construction Scenario Overview

1212 Lobdel Avenue is a 1.39-acre cleared infill site within the City of Wilmington, owned by the Wilmington Housing Authority. The site is zoned R-3 (One-Family Row Houses, Chapter 48) and is targeted for 30+ units of affordable for-sale housing. M4D's predevelopment analysis evaluated four construction systems and three fee-simple yield configurations to give WHA a full-spectrum cost and quality comparison.

2.1 Construction System Comparison (36-Unit Baseline)

	Scn A CLT/SIP-Hybrid	Scn B Off-Site Modular	Scn C Stick-Build	Scn D Garage-Base / Garden Roof
Total Units	36	36	36	36
Unit Configuration	18 ft × 35 ft, 2-story	18 ft × 35 ft, 2-story	18 ft × 35 ft, 2-story	18 ft × 35 ft, 3-level Garage + L2 + L3 + Garden Roof
Unit Size	1,260 SF	1,260 SF	1,260 SF	1,890 SF (living)
Total ROM	\$14,875,352	\$9,200,026	\$14,731,123	\$11,123,852
Cost / Unit	\$413,204	\$255,556	\$409,198	\$308,995
Gap vs \$215K	~\$198K	~\$41K	~\$194K	~\$94K
Parking Strategy	Shared rear court	Shared rear court	Shared rear court	Private garage per unit
Amenity Zone	112.6 ft surplus	112.6 ft surplus	112.6 ft surplus	112.6 ft — full community green

CLT/SIP-Hybrid — Campus Scenario Context

The CLT/SIP-Hybrid scenario (Scenario A) reflects current Mid-Atlantic market conditions including a \$8–15/SF freight premium on CLT panels sourced from out-of-region suppliers and a Florida supply chain for AAC party wall panels. The Merge4Ward regional manufacturing campus — currently in early development with Wilmington-area project partners — is designed to eliminate both supply chain constraints. When operational, the campus brings the CLT/SIP-Hybrid cost into near-parity with Scenario B (off-site modular), while delivering a noncombustible, energy-efficient, regionally manufactured building system. WHA's engagement on this project positions it as the anchor client for the campus's first production run.



MERGE 4 DESIGN LLC

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

2.2 Fee-Simple Yield Options — WHA Homeownership Program

Yield Option	Units	Total ROM (Scn B Modular)	Key Condition
Option B — Full independent setbacks 4 E-W rows, N-S facing units	36	\$9,254,826 \$257,078/unit · gap ~\$42K	Standard recorded plat · No variance required
Opt A+FW12 — E-W rows + cross street 33 units: 9 Lobdel-facing + 2×12 E-W	33	\$8,200,524 \$248,501/unit · gap ~\$34K	15 ft rear yard · Cross street as rear boundary · Pre-app confirmation required
Option D — Shared drive easements 6 E-W rows, N-S facing units — Maximum yield	54	\$12,502,724 \$231,532/unit · gap ~\$17K	Drive aisles as rear yard easements · HOA required · City rear yard interpretation confirmation required

Fee-simple lot ownership across all options.

All three yield options are structured for fee-simple individual lot ownership with standard deeds — the WHA preference for first-time homebuyer FHA/conventional financing. Options B and D require pre-application confirmation of the rear yard easement interpretation with the City of Wilmington Land Use Division. Option A+FW12 additionally requires confirmation of the 15 ft rear yard and cross street as a valid rear yard boundary. All pre-application questions are identified in the Recommended Next Steps section.



MERGE 4 DESIGN LLC

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

03. Scope of Services

M4D proposes services in two tracks. Track A — Predevelopment Intelligence — is authorized and delivered as a standalone fixed-fee engagement. Track B — Architecture & Engineering — is authorized separately upon WHA's selection of a construction type and yield option.

3.1 Track A — Predevelopment Intelligence (Fixed Fee)

Item	Deliverable	Format	Schedule
A1	Site Assessment — parcel confirmation, governing zoning analysis, applicable code standards	Memo / Data Table	Week 1
A2	Yield Scenario Analysis — fee-simple lot configurations and unit counts for Options B, A+FW12, and D; regulatory envelope confirmation	Report + Site Diagrams	Week 1
A3	Construction Type Matrix — four systems (A through D) analyzed across all yield options with cost ranges, per-unit cost, and gap analysis	Matrix Report	Week 1–2
A4	Sustainability Add-Alternate Analysis — geothermal HVAC (shared bore field), community solar PV aisle canopy, and rainwater harvesting; PACE financing and virtual net metering applicability	Analysis Memo	Week 2
A5	3D Site Yield Study & Visualization — massing studies for all yield options; solar access analysis; site plan photorealistic perspectives	3D Visualizations + Site Plan	Week 2–3
A6	WHA Presentation Package — yield option comparison, construction cost matrix, sustainability options, and subsidy gap analysis (PPTX + PDF)	Presentation Deck	Week 3–4
A7	Subsidy Gap & Grant Program Memo — Delaware Catalyst Fund, HOME, and DSHA program fit analysis	Memo	Week 3
A8	Gate 0 ROM Lock — cost estimates reconciled and locked	Summary	Week 4

3.2 Track B — Architecture & Engineering Services (Lump Sum by Phase)

Track B is authorized separately following WHA's written selection of yield option and construction type. Fees below based on the 36-unit baseline; Yield Option D (54 units) will be subject to a fee addendum reflecting the additional design scope.

**MERGE 4 DESIGN LLC**

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

Phase	Scope	Basis	Fee
B1	Schematic Design (SD) — selected yield option; unit plan types; 3D visualizations; site plan	Lump Sum	\$11,700
B2	Design Development (DD) — system coordination; structural and panel coordination	Lump Sum	\$17,550
B3	Construction Documents (CD) — permit-ready stamped set	Lump Sum	\$35,100
B4	Permitting & AHJ Coordination — City of Wilmington permit submission; plan review response	Hourly NTE	\$5,250
B5	Construction Administration (CA) — site observation; RFI/submittal review; punch list	Hourly NTE	\$9,450
B6	Reimbursable Expenses — printing, travel, plots (billed at cost)	Actual	~\$4,000

Subconsultants (Civil Engineer, Structural Engineer, MEP Engineer) are not included in the M4D fee. Budget approximately 12% of construction cost. Subconsultants will be engaged per written authorization from WHA and M4D.

**MERGE 4 DESIGN LLC**

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726**04. Fee Summary****4.1 Track A — Predevelopment Intelligence (Fixed Fee)**

Item	Service	Hrs (Est.)	Rate	Fee
A1	Site Assessment & Zoning Analysis	18	\$185/hr	\$3,330
A2	Yield Scenario Analysis + Site Diagrams	24	\$185/hr	\$4,440
A3	Construction Type Matrix (4 systems × 3 yield options)	8	\$185/hr	\$1,480
A4	Sustainability Add-Alternate Analysis	6	\$185/hr	\$1,110
A5	3D Site Yield Study & Visualization	12	\$165/hr	\$1,980
A6	WHA Presentation Package	10	\$165/hr	\$1,650
A7	Subsidy Gap & Grant Program Memo	8	\$165/hr	\$1,320
A8	Gate 0 ROM Lock	5	\$185/hr	\$925
TRACK A TOTAL — FIXED FEE		91 hrs	—	\$16,235

4.2 Track B — Architecture & Engineering (Lump Sum by Phase)

Phase	Scope	Basis	Fee
B1	Schematic Design	Lump Sum	\$11,700
B2	Design Development	Lump Sum	\$17,550
B3	Construction Documents	Lump Sum	\$35,100
B4	Permitting & AHJ Coordination	Hourly NTE	\$5,250
B5	Construction Administration	Hourly NTE	\$9,450
B6	Reimbursable Expenses	Actual	~\$4,000
TRACK B SUBTOTAL		—	\$83,050

4.3 Combined Fee & Track A Credit

	Amount
Track A — Predevelopment Intelligence (Fixed Fee)	\$16,235
Track B — Architecture & Engineering (Lump Sum by Phase)	\$83,050
Track B — Reimbursable Expenses (estimated)	~\$4,000
COMBINED TOTAL — TRACKS A + B	~\$103,285



MERGE 4 DESIGN LLC

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

★ **TRACK A FEE CREDIT** — Upon WHA's written authorization of Track B (full design services through Construction Documents), M4D will credit the full Track A fee of \$16,235 against the Track B Phase B1 invoice. This credit reflects M4D's commitment to a long-term WHA relationship and its confidence in this project proceeding to full design authorization.

(\$16,235)

NET TOTAL AFTER TRACK A CREDIT (upon Track B authorization)

~\$87,050



MERGE 4 DESIGN LLC

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

05. Recommended Next Steps

#	Action	Lead	Timeline
1	Authorize Track A — Predevelopment Intelligence (\$16,235 fixed). Execute M4D PSA M2_2026_005.	WHA + M4D	Immediate
2	Request City of Wilmington Land Use pre-application meeting. Agenda: R-3 use confirmation; block face setback average; 10-unit row fire code; Scenario D height compliance (Sec. 48-151); fee-simple lot yield options — rear yard easement interpretation for Option D (shared drive aisles as rear yard), cross street as rear boundary for Option A+FW12, and 15 ft rear yard variance feasibility.	M4D	Week 1
3	Order Phase I Environmental Site Assessment — prior school facility history; possible asbestos, underground tanks, contaminated soil.	WHA	Week 1
4	Authorize geotechnical investigation (minimum 4 test borings). Critical for foundation design and geothermal bore field feasibility.	WHA	Week 1–2
5	Confirm WHA homebuyer income targeting (% AMI) and subordinate financing capacity — determines subsidy stack and Catalyst Fund eligibility.	WHA Finance	Week 2
6	RFI to modular manufacturers: 18 ft wide × 35 ft deep × 2-story, 36–54 units, Mid-Atlantic delivery. Confirm Scenario D podium-set capability.	M4D	Week 2
7	Confirm Delaware Catalyst Fund eligibility pathway. Coordinate with DSHA.	M4D + WHA	Week 2–3
8	Confirm Energize Delaware PACE eligibility for individual fee-simple lots; confirm Delmarva Power virtual net metering participation for HOA subscriber structure.	M4D	Week 2–3
9	WHA presentation — construction type options, yield configurations, ROM comparison, sustainability systems.	M4D + WHA	Week 4–5
10	WHA selection of yield option and construction type; written authorization of Track B. Track A credit applied at Phase B1 invoice.	WHA	Post-presentation



MERGE 4 DESIGN LLC

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

06. Authorization — Track A

To authorize Track A services, please execute this proposal and return a signed copy to M4D. A Professional Services Agreement will be issued upon receipt. Services commence within three (3) business days of executed PSA and receipt of retainer.

Track A Fixed Fee

\$16,235.00

Retainer (50% due upon authorization): \$8,116.50

Delivery: 4 weeks from written authorization

Governing agreement: M4D PSA
M2_2026_012-PSA (to be issued)

Invoice: Net 15 days

AUTHORIZED BY — MERGE 4 DESIGN LLC

Trevor Knight, Dir. of Business Development

David Ainsworth AIA, Principal-in-Charge

Date: _____

AUTHORIZED BY — WHA

Name: _____

Title: _____

Date: _____

FLAG FOR LEGAL/FINANCIAL REVIEW: Confirm WHA procurement requirements (DBE/MBE thresholds, HUD Section 3, City of Wilmington local hiring provisions) before contract execution. This proposal does not constitute a contract. ROM figures do not constitute a bid, guarantee, or warranty of construction cost. Professional legal review is recommended prior to execution.



MERGE 4 DESIGN LLC

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

07. Rough Order of Magnitude Cost Analysis — All Four Construction Systems — DRAFT

DRAFT — FOR INTERNAL REVIEW ONLY — NOT FOR DISTRIBUTION

Cost figures in this section are Rough Orders of Magnitude for internal planning purposes. These estimates are subject to revision and do not constitute a bid, guarantee, or warranty of construction cost. Final cost analysis will be issued upon WHA's selection of yield option and construction type.

The following cost estimates are Rough Orders of Magnitude (ROM) based on available site information, current regional labor and material market conditions, and M4D's predevelopment cost analysis. These figures do not constitute a bid, guarantee, or warranty of construction cost. All values are subject to revision upon receipt of Phase I ESA, geotechnical data, confirmed zoning compliance, and construction type selection.

ROM breakdowns below are presented across three unit-count scenarios: Scenario I (36 units), Scenario II (40 units), and Scenario III (24 units). For yield options exceeding 36 units (Options A+FW12 and D), cost scaling is addressed in Section 7.5 Master Comparison Matrix.

7.1 Scenario A — CLT/SIP-Hybrid

**Cross-laminated timber structural panels + structurally insulated envelope panels. Off-site panel fabrication; on-site erection. Current Mid-Atlantic freight premium applies (\$8–15/SF). Cost position improves materially when regional manufacturing supply chain is established.*

Line Item	Scn I — 36 Units	Scn II — 40 Units	Scn III — 24 Units
Gross Floor Area (2-story)	45,360 SF	44,800 SF	36,480 SF
Sitework / Civil	\$432,000	\$480,000	\$288,000
Foundations (CLT-spec bearing, party wall footings)	\$1,179,360	\$1,164,800	\$948,480
CLT Structural Panels (supply + freight premium)	\$2,630,880	\$2,598,400	\$2,115,840
SIP Envelope Panels (ext walls + roof)	\$907,200	\$896,000	\$729,600
Erection Crew + Crane	\$360,000	\$400,000	\$240,000
Rough-in MEP	\$1,360,800	\$1,344,000	\$1,094,400
Interior Finishes	\$1,905,120	\$1,881,600	\$1,531,360
Doors, Windows, Hardware	\$270,000	\$300,000	\$180,000
Appliances Allowance	\$144,000	\$160,000	\$96,000
Site Utilities	\$306,000	\$340,000	\$204,000
Shared Open Space / Parking Court	\$200,000	\$200,000	\$175,000
Landscaping / Fencing / Unit Walks	\$162,000	\$180,000	\$108,000

**MERGE 4 DESIGN LLC**

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

SUBTOTAL — BASE CONSTRUCTION	\$9,857,360	\$9,944,800	\$7,710,680
General Conditions (10%)	\$985,736	\$994,480	\$771,068
OH&P / GC Fee (8%)	\$868,648	\$875,942	\$679,341
Design Contingency (8%)	\$937,580	\$945,458	\$733,449
Soft Costs — Arch, Eng, Permits, Testing (12%)	\$1,517,678	\$1,530,742	\$1,187,447
Owner Contingency (5%)	\$708,350	\$714,571	\$554,099
SCENARIO A — TOTAL ROM	\$14,875,352	\$15,005,993	\$11,635,084
Cost Per Unit	\$413,204	\$375,150	\$484,795
Gap vs \$215K Target	~\$198K/unit	~\$160K/unit	~\$270K/unit



MERGE 4 DESIGN LLC

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

3.2 Scenario B — Off-Site Modular

**Factory-built volumetric modules, crane-set on-site. Fastest schedule, highest quality control, lowest subsidy gap among the four systems in current market conditions.*

Line Item	Scn I — 36 Units	Scn II — 40 Units	Scn III — 24 Units
Gross Floor Area (2-story)	45,360 SF	44,800 SF	36,480 SF
Sitework / Civil	\$432,000	\$480,000	\$288,000
Foundations (modular-spec slab/piers)	\$816,480	\$806,400	\$656,640
Modular Factory Units (factory-complete, MEP rough-in incl.)	\$3,009,600	\$3,344,000	\$2,016,000
Freight + Set (crane, transport)	\$540,000	\$600,000	\$360,000
Site MEP Tie-Ins + Utility Connections	\$306,000	\$340,000	\$204,000
Exterior Finish Completion (siding, roofing, trim)	\$726,000	\$716,800	\$583,680
Interior Finish Completion (punch-out items)	\$234,000	\$260,000	\$156,000
Shared Open Space / Parking Court	\$200,000	\$200,000	\$175,000
Landscaping / Fencing / Unit Walks	\$162,000	\$180,000	\$108,000
SUBTOTAL — BASE CONSTRUCTION	\$6,426,080	\$6,927,200	\$4,547,320
General Conditions (8%)	\$514,086	\$554,176	\$363,786
OH&P / GC Fee (6%)	\$416,770	\$449,483	\$294,968
Design Contingency (6%)	\$441,776	\$476,452	\$312,606
Soft Costs — Arch, Eng, Permits (12%)	\$963,217	\$1,038,759	\$681,321
Owner Contingency (5%)	\$438,097	\$472,303	\$309,700
SCENARIO B — TOTAL ROM	\$9,200,026	\$9,918,373	\$6,509,701
Cost Per Unit	\$255,556	\$247,959	\$271,238
Gap vs \$215K Target	~\$41K/unit	~\$33K/unit	~\$56K/unit



MERGE 4 DESIGN LLC

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

3.3 Scenario C — Conventional Stick-Build

**Traditional wood-frame construction. Highest local GC availability; longest schedule; highest labor content. Retained as baseline benchmark for grant applications requiring conventional cost comparison.*

Line Item	Scn I — 36 Units	Scn II — 40 Units	Scn III — 24 Units
Gross Floor Area (2-story)	45,360 SF	44,800 SF	36,480 SF
Sitework / Civil	\$432,000	\$480,000	\$288,000
Foundations (slab-on-grade, party wall footings)	\$816,480	\$806,400	\$656,640
Framing Lumber + Labor	\$1,587,600	\$1,568,000	\$1,276,800
Sheathing + Insulation + Weather Barrier	\$589,680	\$582,400	\$474,240
Roofing	\$363,480	\$358,400	\$291,840
Exterior Finish (fiber cement + brick base)	\$907,200	\$896,000	\$729,600
Rough MEP	\$1,587,600	\$1,568,000	\$1,276,800
Insulation + Drywall	\$635,040	\$627,200	\$510,720
Interior Finishes	\$1,724,688	\$1,702,880	\$1,386,240
Doors, Hardware, Appliances	\$414,000	\$460,000	\$252,000
Site Utilities + Connections	\$306,000	\$340,000	\$204,000
Shared Open Space / Parking Court	\$200,000	\$200,000	\$175,000
Landscaping / Fencing / Unit Walks	\$162,000	\$180,000	\$108,000
SUBTOTAL — BASE CONSTRUCTION	\$9,725,768	\$9,769,280	\$7,630,080
General Conditions (10%)	\$972,577	\$976,928	\$763,008
OH&P / GC Fee (8%)	\$855,868	\$859,697	\$671,847
Design Contingency (10%)	\$1,055,421	\$1,060,591	\$826,694
Soft Costs (12%)	\$1,448,578	\$1,455,011	\$1,134,194
Owner Contingency (5%)	\$672,911	\$676,076	\$527,292
SCENARIO C — TOTAL ROM	\$14,731,123	\$14,797,583	\$11,553,115
Cost Per Unit	\$409,198	\$369,939	\$481,380
Gap vs \$215K Target	~\$194K/unit	~\$155K/unit	~\$266K/unit



MERGE 4 DESIGN LLC

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

3.4 Scenario D — Garage-Base / Garden Roof Townhouse (36 Units)

**Scenario D: Level 1 CMU/light steel garage on slab-on-grade. Levels 2–3 off-site modular living stack. Roof: usable garden/green roof deck. Living program 1,890 SF per unit (3BR/2.5BA). Private garage per unit; amenity zone reprogrammed as full community green. Height compliance with Chapter 48 Sec. 48-151 (35 ft max) to be confirmed at City pre-application meeting.*

Line Item	Scenario I — 36 Units
Living GFA (L2 + L3, 2-story modular stack)	68,040 SF (36 × 1,890 SF)
Garage GFA (L1, CMU/steel slab-on-grade)	22,680 SF (36 × 630 SF)
Sitework / Civil	\$432,000
Foundations — Slab-on-Grade	\$756,000
Garage Level Shell — CMU/light steel, 36 units @ \$22,000	\$792,000
Modular Factory Units — L2+L3 living stack, MEP rough-in incl.	\$3,009,600
Freight + Set — crane lift over garage podium	\$580,000
Site MEP Tie-Ins + Utility Connections	\$306,000
Exterior Finish Completion — 3-level facade	\$780,000
Interior Finish Completion + Module-to-Garage Interface	\$280,000
Garden Roof Deck — usable, structural, waterproofed, 36 units @ \$24,000	\$864,000
Amenity Zone / Community Green (visitor parking only)	\$120,000
Landscaping / Fencing / Unit Walks	\$200,000
General Conditions (8%)	\$649,568
OH&P / GC Fee (6%)	\$487,176
Design Contingency (6%)	\$487,176
Soft Costs — Arch, Eng, Permits (12%)	\$974,352
Owner Contingency (5%)	\$405,980
SCENARIO D — TOTAL ROM	\$11,123,852
Cost Per Unit	\$308,995
Gap vs \$215K Target	~\$94K/unit · ~\$3,383,820 total

3.5 Master Comparison Matrix

Scenario / Yield Option	Units	Total ROM	\$/Unit	Gap/Unit	Rating
A — CLT/SIP-Hybrid (36 units)	36	\$14,875,352	\$413,204	~\$198K	C



MERGE 4 DESIGN LLC

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

B — Off-Site Modular (36 units)	36	\$9,200,026	\$255,556	~\$41K	A+
C — Stick-Build (36 units)	36	\$14,731,123	\$409,198	~\$194K	C
D — Garage-Base / Garden Roof (36 units)	36	\$11,123,852	\$308,995	~\$94K	B
B — Off-Site Modular (40 units, Scn II)	40	\$9,918,373	\$247,959	~\$33K	A++
B — Off-Site Modular (24 units, Scn III)	24	\$6,509,701	\$271,238	~\$56K	B
— FEE-SIMPLE YIELD OPTIONS (modular construction) —					
Yield Opt B — 36 units, full indep. setbacks	36	\$9,254,826	\$257,078	~\$42K	A+
Yield Opt A+FW12 — 33 units, cross street + fire wall	33	\$8,200,524	\$248,501	~\$34K	A
Yield Opt D — 54 units, shared easements ★ Max yield	54	\$12,502,724	\$231,532	~\$17K	A++

Rating methodology: A++ = gap closeable with Delaware Catalyst Fund alone (\$35K/unit × 54 = \$1.89M available vs ~\$917K needed); A+ = Catalyst Fund covers ~70%; A = Catalyst Fund + modest HOME supplement; B = layered subsidy required; C = extraordinary subsidy; D = not viable under current financing structure.

3.6 Sustainability Systems Add-Alternate

The following systems are available as add-alternates to any construction scenario and yield option. All costs and savings are indexed to the 54-unit fee-simple yield (Option D) and scaled for other options in the table below. All three systems are designed to be installed under private drive aisles or unit rooftops with no impact on lot boundaries or amenity zones.

System	Total Cost	Per Unit	Key Feature
Geothermal HVAC — shared bore field (37 bores, drilled under drive aisles)	\$672,217	\$12,448/unit	4.4-yr simple payback · \$75/unit/mo energy savings · PACE eligible
Community Solar PV — aisle canopy (191 kW, 267,120 kWh/yr, 66% of community energy)	\$599,040 gross \$419,328 net after 30% ITC	\$7,765/unit (net)	Virtual net metering · \$58/unit/mo savings · Covered parking



MERGE 4 DESIGN LLC

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

Rainwater Harvesting — 6,000-gal underground cistern, irrigation only	\$26,820	\$497/unit	Stormwater management credit · Enterprise Green Communities points
---	----------	------------	--

HOA Shared Maintenance Fee	Annual	Monthly / Unit
Geothermal loop (pump, glycol, controls)	\$2,400	\$3.70
Solar PV (inverters, monitoring, cleaning)	\$3,200	\$4.94
Rainwater system (pump, filters, inspection)	\$1,200	\$1.85
Drive aisle maintenance (seal coat, snow removal)	\$4,000	\$6.17
Landscaping + irrigation	\$8,500	\$13.12
Shared insurance	\$3,600	\$5.56
Reserve fund (10%) + management	\$4,690	\$7.24
TOTAL HOA BUDGET	\$27,590/yr	\$43/unit/mo

Monthly Cash Flow per Unit (all 3 systems)	Out	Savings
PACE payment — geothermal premium (20yr, 5.5%)	\$28/mo	—
PACE payment — solar net of 30% ITC (20yr, 5.5%)	\$54/mo	—
HOA monthly fee (all shared systems)	\$43/mo	—
Energy savings — geothermal	—	\$75/mo
Solar net metering credits (Delmarva Power)	—	\$58/mo
Water savings — irrigation	—	\$1/mo
NET MONTHLY PER UNIT	\$125/mo total out	\$134/mo total savings

★ **Savings exceed cost on day one:** +\$9/unit/month net — energy and solar savings outpace PACE payments and HOA fee from day one. At 80% AMI (~\$68K/yr household income), total PITI + HOA = \$1,652/mo, qualifying for FHA financing without income limitation.

Sustainability system costs indexed to 54-unit yield. Scale factors: geo costs scale with units; solar canopy and rainwater system are largely fixed to the site. HOA fee reduces to ~\$45/unit/mo at 36 units. Refer to the M4D Sustainability Add-Alternate Analysis (Deliverable A4) for full system specifications, PACE structure, and Delaware virtual net metering confirmation process.



MERGE 4 DESIGN LLC

ARCHITECTURE DEVELOPMENT CONSULTING

500 Delaware Ave., Suite 1 #1551,
Wilmington, DE 19899 · info@merge4design.com · 302 367 5726

- △ Confirm with Energize Delaware that PACE attaches to individual fee-simple lots.
- △ Confirm Delmarva Power participation in virtual net metering for HOA subscriber structure.
- △ HOA covenant must address geothermal loop access rights, solar canopy maintenance, and aisle easement language. Attorney review required before plat recording.